



$$16^{x-9} = 0,5 = \frac{1}{2} = 2^{-1}$$

$$16^{x-9} = 2^{-1}$$

$$16 = 2^4$$

$$2^{4(x-9)} = 2^{-1}$$

$$4(x-9) = -1$$

$$4x - 36 = -1$$

$$4x = 35$$

$$x = \frac{35}{4}$$

$$81^{3-x} = \frac{1}{9} = 9^{-1}$$

$$9^{2(3-x)} = 9^{-1}$$

$$2(3-x) = -1$$

$$2x = 6 + 1 = 7$$

$$2x = 7$$

$$x = \frac{7}{2} = 3,5$$

$$(0,125)^{x-2} = 64$$

$$\frac{125}{1000} = \frac{25}{200} = \frac{5}{40} = \frac{1}{8}$$

$$\left(\frac{1}{8}\right)^{x-2} = 64$$

$$(8^{-1})^{(x-2)} = 64 = 8^2$$

$$-(x-2) = 2$$

$$-x + 2 = 2$$

$$x = 0$$

$$\log_3(6+5x) = \log_3(4-5x) + \underline{\underline{2}}$$

$$2 = 2 \cdot \log_3 3 = \log_3 3^2$$

$$\log_3 (6+5x) = \log_3 (4-5x) + \log_3 9$$

$$\log_a b + \log_a c = \log_a b \cdot c$$

~~$$\log_3 (6+5x) = \log_3 (9(4-5x))$$~~

$$6+5x = 9(4-5x)$$

$$6+5x = 36 - 45x$$

$$50x = 30$$

$$x = \frac{30}{50} = \frac{3}{5} = 0.6$$

$$\log_2 (8+3x) = \log_2 (3+x) + \frac{1}{11} \log_2 2$$

$$x = -2$$

$$\log_4 (5+6x) = \log_4 (3+4x) + 3 \log_4 4$$

$$5+6x =$$

$$\log_4 4^3$$

$$3 \cdot \underbrace{1}_{\log_4 4} = 3 \cdot \log_4 4 = \log_4 4^3 =$$

$$\log_4 4$$

$$\log_4 64$$

$$\log_a b^c = c \log_a b$$

$$\log_4 (5+6x) = \log_4 (3+4x) + \log_4 64$$

$$5+6x = 64(3+4x)$$

$$\log_5 (2+x) = \log_5 (2x+1) + 2$$

$$\log_5(2+x) = \log_5(2x+1) + \log_5 25$$

$$2+x = 25(2x+1)$$

$$2+x = 50x+25$$

$$50x - x = 2 - 25$$

$$49x = -23$$

$$x = \frac{-23}{49}$$

Неприведенная

$$\frac{A}{B} \geq 0$$

$$A = (x-2)(x+3)$$

1. ищем корни A

$$x=2$$

$$x=-3$$

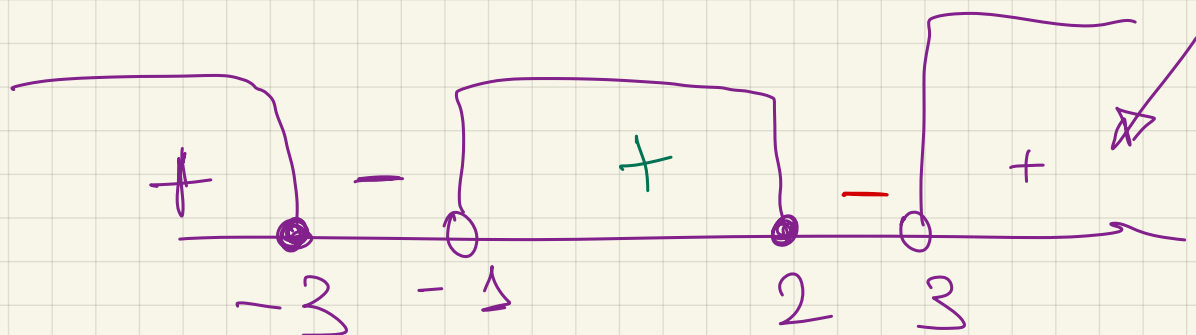
$$B = (x+1)(x-3)$$

2. ищем корни B

$$x=-1$$

$$x=3$$

3. Bei der folgenden Ungleichung:



$$\frac{(x-2)(x+3)}{(x+1)(x-3)} \leq 0$$

$$x \in (-\infty; -3] \cup (-1; 2] \cup (3; +\infty)$$

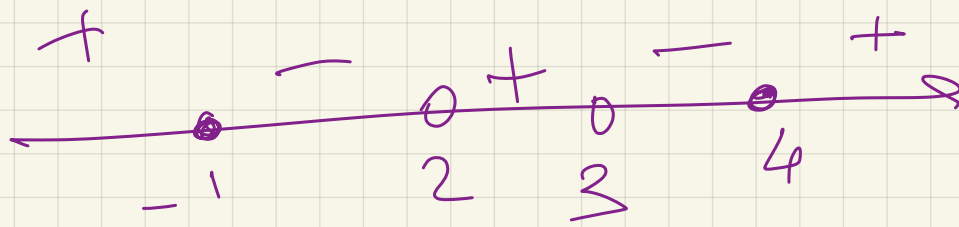
$$\frac{(x+1)(x-4)}{(x-3)(x-2)} \leq 0$$

1. Nullstellen des Zähler:

$$\begin{array}{ll} x+1=0 & x=-1 \\ x-4=0 & x=4 \end{array}$$

2. Nullstellen des Nenner:

$$\begin{array}{l} x=3 \\ x=2 \end{array}$$



$$x \in [-1; 2) \cup (3; 4]$$

$$\frac{(x-2)^2}{(x-1)(x-5)} \geq 0$$

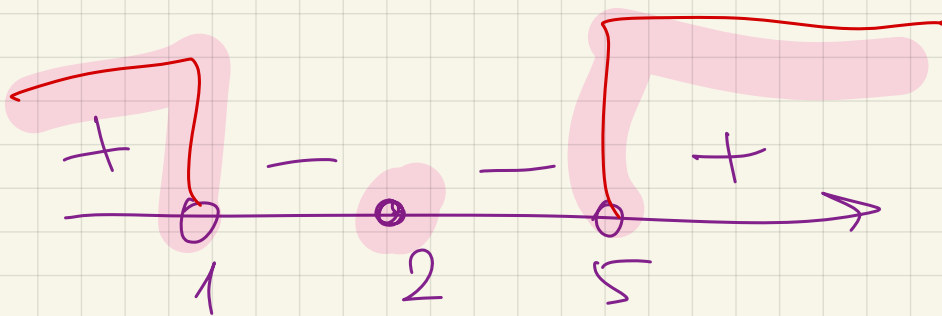
$$(x-2)^2 = 0$$

1. $x = 2$

$$x^2 - 2 \cdot 2x + 4 = x^2 - 4x + 4$$

2. $x = 1$
 $x = 5$

$$(a \cdot b)^2 = a^2 - \underline{\underline{2ab}} + b^2$$



$$x \in (-\infty; 1) \cup (5; +\infty) \cup \{2\}$$

$$\frac{(x-1)(x-5)}{(x-2)^2} \geq 0$$